Three questions, three different levels of evidence

Purpose

Do the five researcher-defined categories have different spectra? Can we recognize an unseen observation? Can we recover category fractions from a mixture? These questions demand progressively stronger evidence. A positive answer at one level is not a positive answer at the next.

The biological grouping

Deciduous; Evergreen; Herbaceous; Non-vegetated & dead; Woody shrub. Only the recorded cover_category field enters this primary analysis. Historical groupings and all earlier reports remain preserved separately.

The evidence progression

Spectra can differ → differences may allow category recognition → recognized components may or may not have recoverable abundance → real field cover still requires independent validation.

The main result

All ten category pairs differ in the sampled blocked comparisons. Full-NEON recognition BA is 85.2% for unseen polygons and 73.8% for held sites. Synthetic ridge120 fraction RMSE is 0.216/0.269, but the same-candidate ridge15 control loses that advantage, and global-year ridge does not beat the no-spectrum prior.

What changed after the audit

This is more than a rearranged report: five random-subset paths per budget, PCA recognition controls, a consistent nested temporal classifier, a 15-parent same-candidate ridge test, class-level recognition/abundance links and explicit adverse explanations were added. Valid prior evidence was reused.

Scope

This is a structured reanalysis of an extensively explored observational library, not an independently preregistered confirmation. Scores concern labelled polygon footprints and synthetic algebraic weights. No independent field fractional-cover validation exists.

Evidence: tables/claim_registry.csv; DESIGN.md; ABUNDANCE_CONTROL_DESIGN.md

Foundational validation: observations, support and independence

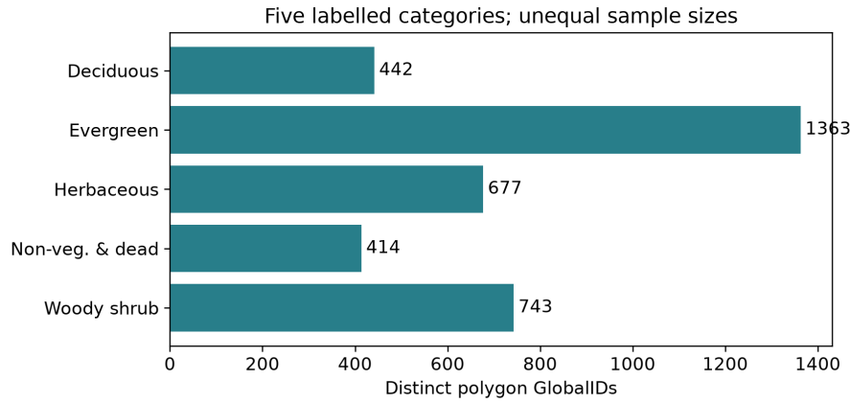
Source validity

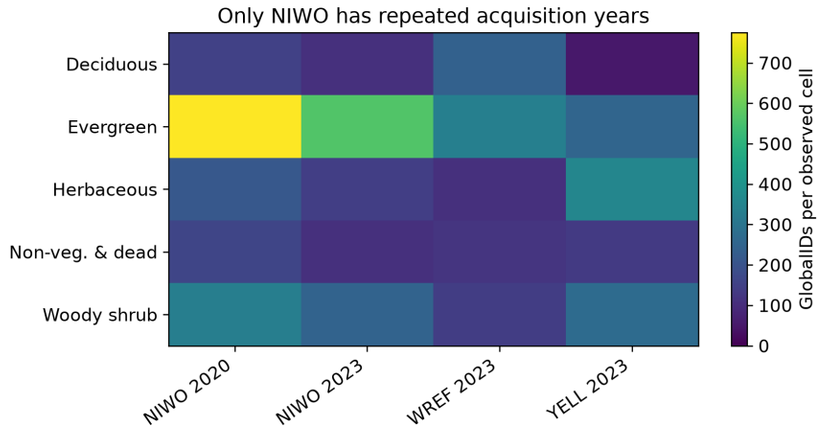
The supplied table has 85,706 pixel rows. Its merged corrected schema contains 1,140 native wavelength columns because source products have different native coordinates; each pixel has exactly 426 finite corrected native values. Structural null padding must not be called analytical-grid missingness. The retained 360-coordinate polygon/year grid is finite, and no exact duplicate annual NEON/OLI spectra were found.

Truth and units

Annotations date to 2023/2024 and are reused across acquisition years. GlobalID denotes a polygon footprint, not certified plant/material purity. Absolute stored reflectance scaling is unresolved; no guessed correction, rescaling or interpolation was introduced.

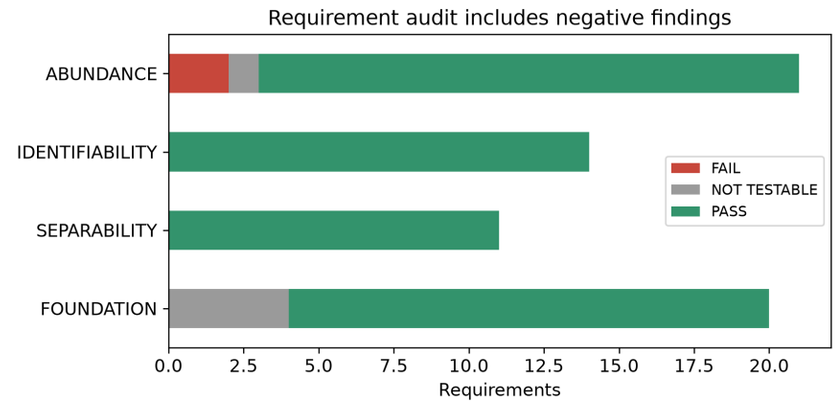
The biological sampling unit is the polygon, not the pixel





85,706 extracted pixel records
↓ summarize within footprint/acquisition
4,807 polygon/year spectra
↓ cluster repeated acquisitions
3,639 independent polygon IDs

1,168 IDs observed in both years
Pixels, channels and mixtures are not replicates.



HOW TO READ IT
Counts use distinct GlobalIDs. A repeated ID can occur in two acquisition-year cells. Audit FAIL refers to a tested claim failing, not a concealed pipeline error.

WHAT WE LEARN
The library has broad category coverage but only four site/year cells. The experimental independence and truth limits are explicit.

WHAT REMAINS UNCERTAIN
Polygon purity, absolute reflectance units and independent annual/field fraction truth are not verified. Distinct nearby polygons can remain spatially correlated.

Numerical sources: tables/category_support.csv; tables/site_year_category_support.csv; foundation.json; tables/design_validation.csv

Foundational validation: what passes and what cannot be tested

How to read this audit

PASS means the stated design requirement was met or its limitation was documented by evidence. FAIL means a substantive tested claim failed. NOT TESTABLE means the required truth or independent support is unavailable. A procedurally valid experiment may correctly produce a negative scientific result.

Requirement	Status	Test/result
GlobalID biological sampling unit	PASS	3,639 identities; 4,807 annual spectra; 1,168 repeated IDs
Category sample sizes and imbalance	PASS	All five categories counted by distinct GlobalID, not pixel rows
Site/year coverage	PASS	Four observed site/year cells, not balanced crossed coverage
Repeated identities and split grouping	PASS	All acquisitions of each target ID excluded from model training
Spectral coordinate count	PASS	360 usable NEON coordinates; seven synthetic OLI channels
Atmospheric gaps	PASS	Existing masked grid retained; no gap interpolation in analysis or plotting
Synthetic OLI construction	PASS	Shared corrected-source convolution; exact configured centers/FWHM, rounded export labels
Annotation dates documented	PASS	2023/2024 annotations reused across imagery years
Missing source values	PASS	Each pixel has 426 finite native coordinates; union schema has structural null padding; compact grid finite
Duplicate annual spectra	PASS	No exact duplicate NEON or OLI annual vectors across or within identities
Exact source-pixel leakage	PASS	Prior metadata audit: no shared source pixels across distinct GlobalIDs
Training-only preprocessing	PASS	Scalers/PCA/selectors fitted within appropriate training roles
Nested feature and model selection	PASS	Matching grouped inner folds; C chosen only by inner scores
Mixture parent membership	PASS	Proper/tuning/test parents separated; same target mixtures for all primary models
Dependent mixtures	PASS	Bootstrap resamples whole parent bundles, never independent synthetic rows
Historical labels excluded	PASS	Only source cover_category enters new models; prior grouping history preserved
Absolute reflectance scaling and physical units	NOT TESTABLE	Stored scale unresolved; no guessed rescaling or physical cross-sensor unit claim.
Independent annual biological truth	NOT TESTABLE	Retrospective category labels cannot validate annual ecological change.
Independent field fractions and purity	NOT TESTABLE	No contemporaneous fractional-cover or pure-material truth supplied.
Spatially independent regions beyond sampled sites	NOT TESTABLE	Three sites do not establish new-region geographic generality; spatial autocorrelation remains.

Evidence: foundation.json; tables/source_corrected_validity.csv; tables/duplicate_spectra.csv; tables/design_validation.csv

1. SEPARABILITY

Scientific question

Are the five researcher categories spectrally different, where do differences occur, and are those differences stable across observed domains? This section concerns difference, not classification or fractions.

Relevant method

Reuse 40 exact-acquisition-history matched cohort contrasts: ten category pairs in NIWO 2020, NIWO 2023, WREF 2023 and YELL 2023. Whole-curve category permutations preserve adjacent-wavelength correlation; 9,999 synchronized shuffles per contrast support the existing maximum, integrated and combined tests. Whole-spectrum evidence uses its declared Holm family.

Wavelength evidence

The wavelength map uses within-contrast max-T and Bonferroni across 40 contrasts. Whole-curve null maxima account for correlated channels; 360 wavelengths are never treated as 360 independent biological samples. Full-cohort inferential maps are descriptive and never select predictive features. Separate training-only evidence was previously validated.

Effect versus evidence

Standardized signed effects and width-weighted RMS whole-spectrum effects quantify magnitude. Adjusted permutation probabilities quantify evidence under the exchangeability assumption. Saved null standard errors/envelopes localize departures from that null; they are not confidence intervals for effect size.

Domain and resolution

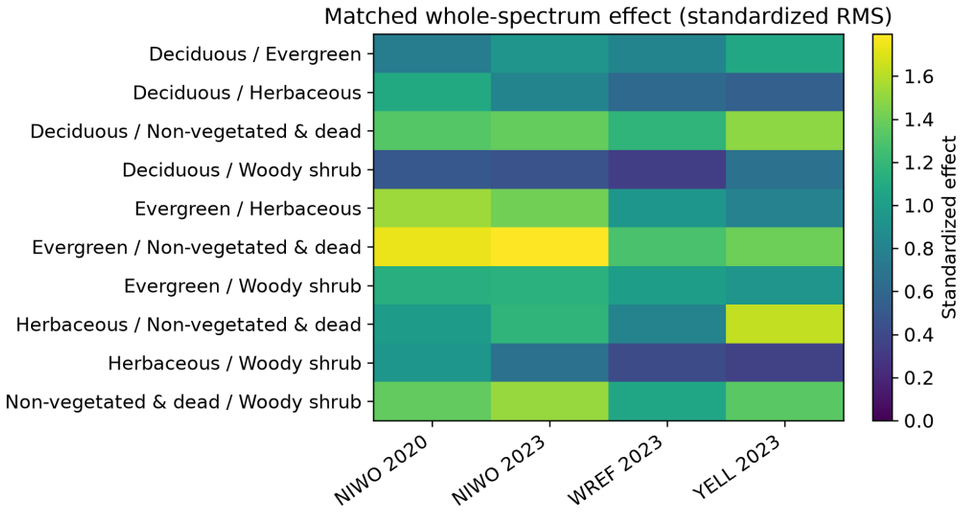
All four cohort effects, wavelength sign/magnitude consistency, rank consistency, weakest observed effect domain and NIWO year differences are retained. Visible, red edge, NIR, SWIR1 and SWIR2 are summaries of disconnected supported coordinates. OLI is a synthetic convolution of the same corrected reference, not independent satellite validation.

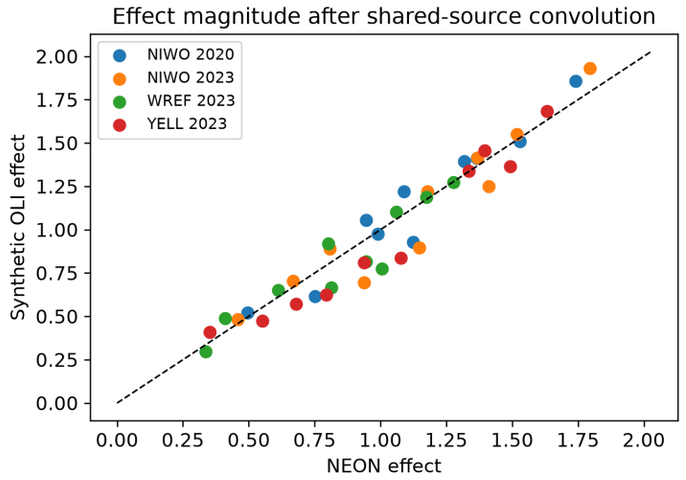
Pair	Cohorts	Minimum class support A/B	Effect range	Largest adjusted p
Deciduous / Evergreen	4	41/73	0.75-1.08	0.004
Deciduous / Herbaceous	4	45/91	0.55-1.09	0.004
Deciduous / Non-vegetated & dead	4	39/41	1.17-1.49	0.004
Deciduous / Woody shrub	4	40/76	0.34-0.68	0.004
Evergreen / Herbaceous	4	169/107	0.79-1.53	0.004
Evergreen / Non-vegetated & dead	4	173/82	1.28-1.79	0.004
Evergreen / Woody shrub	4	199/129	0.94-1.15	0.004
Herbaceous / Non-vegetated & dead	4	82/68	0.80-1.63	0.004
Herbaceous / Woody shrub	4	81/122	0.35-0.95	0.004
Non-vegetated & dead / Woody shrub	4	100/122	1.06-1.52	0.004

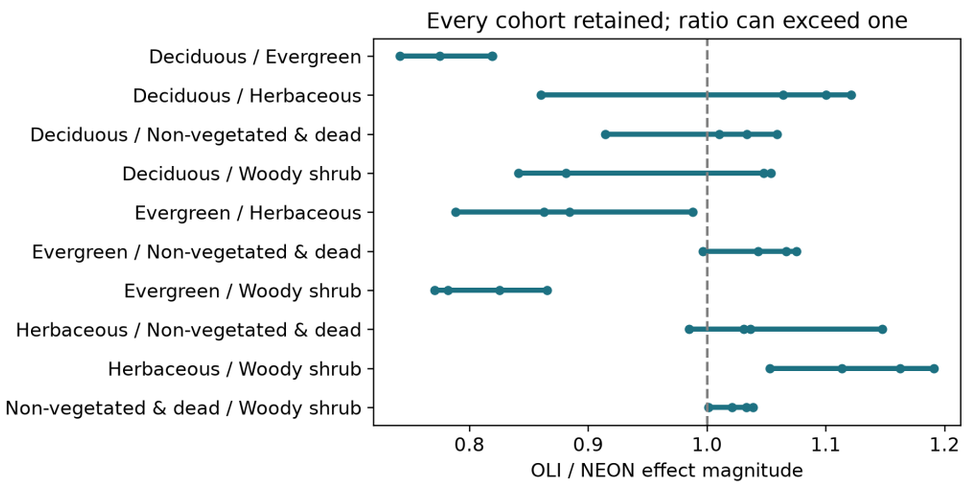
Evidence: tables/pair_support_and_evidence.csv; tables/whole_spectrum_separability.csv; tables/wavelength_effect_stability.csv; tables/wavelength_null_uncertainty.csv; prior:tables/domain_rank_consistency.csv

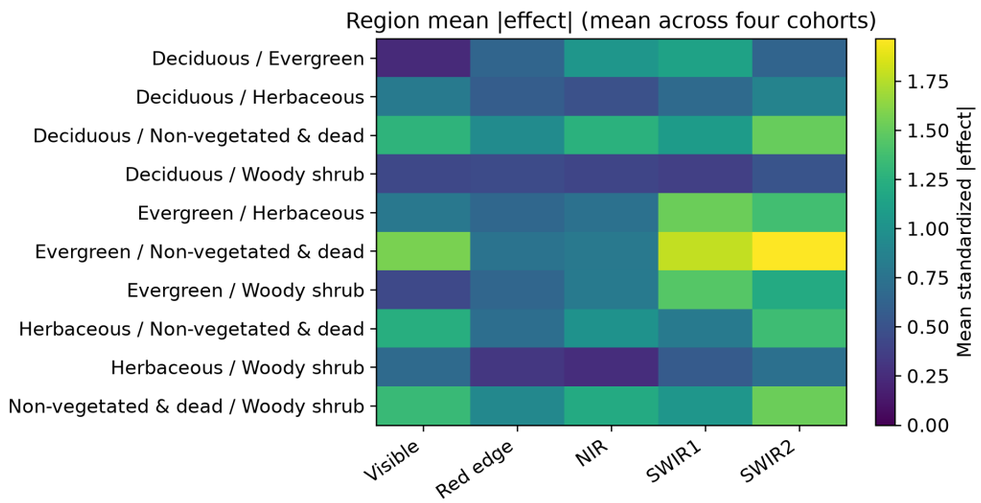
1. SEPARABILITY | effect magnitude and sensor resolution

Different spectra: effect size, statistical evidence and sensor retention









HOW TO READ IT
Each heatmap row is one of ten category pairs. All four cohorts remain visible. Ratios measure standardized effect retention, not prediction accuracy or a probability.

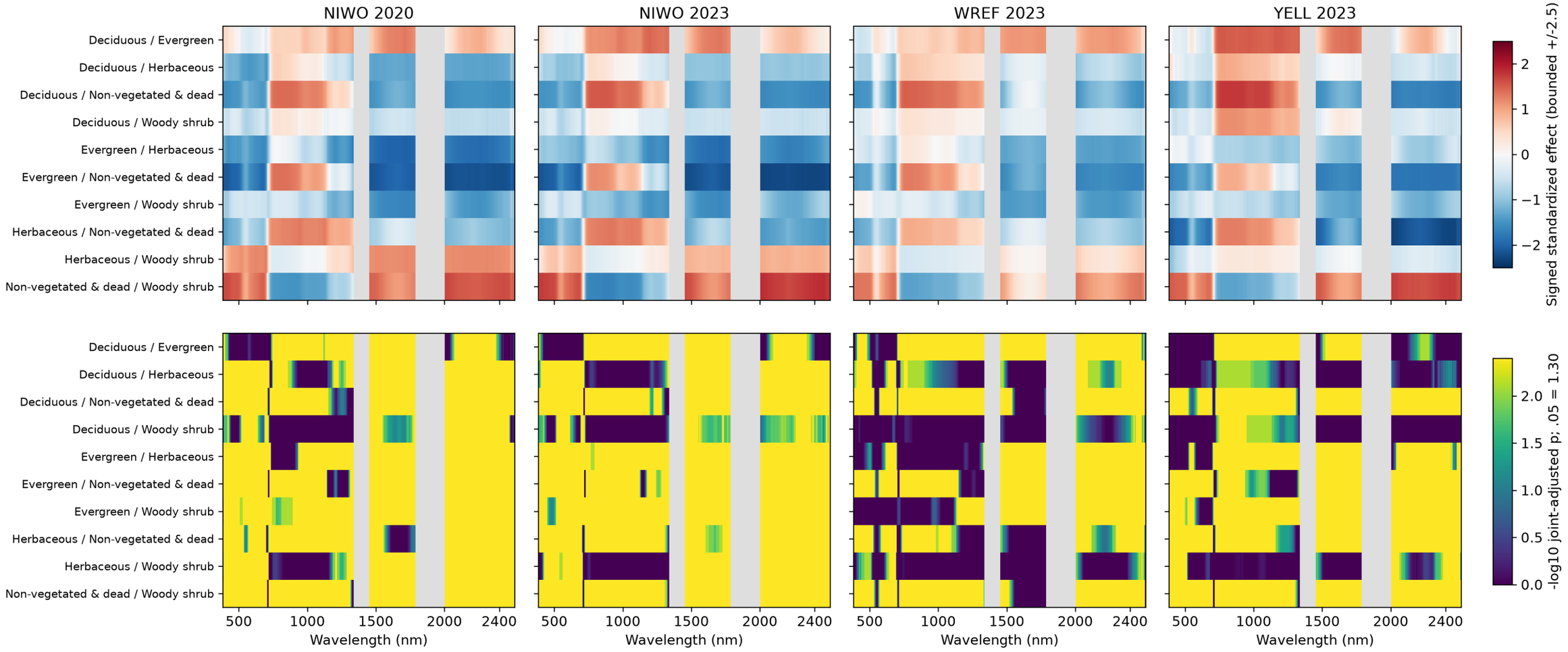
WHAT WE LEARN
All ten whole-spectrum contrasts are supported in the four observed cohorts after declared blocked permutation correction. Effect magnitude and OLI retention vary by pair/domain.

WHAT REMAINS UNCERTAIN
Absolute physical units remain unresolved. Convolution can smooth within-class variability, so a standardized effect ratio above one is possible. Synthetic OLI is not independent Landsat evidence.

Numerical sources: tables/whole_spectrum_separability.csv; tables/pair_support_and_evidence.csv; tables/region_separability.csv

1. SEPARABILITY | every supported wavelength, every observed cohort

Where differences occur, and where they change across domains



HOW TO READ IT

Top: signed matched effect, class A minus B in the displayed pair. Bottom: adjusted evidence; 1.30 corresponds to p=.05. Grey intervals are masked gaps. Color limits bound display only.

WHAT WE LEARN

Many broad regions are supported, but sign/magnitude/rank stability is pair- and domain-dependent. All coordinates share whole-curve permutation shuffles; adjacent channels are not independent tests.

WHAT REMAINS UNCERTAIN

The lower map is evidence against exchangeability, not an effect confidence interval. Full-cohort maps are descriptive and never used to select predictive features. Null standard errors and every unbounded effect are saved.

Numerical sources: prior:tables/cohort_wavelength_inference.csv; tables/wavelength_effect_stability.csv; tables/wavelength_null_uncertainty.csv; prior:tables/domain_rank_consistency.csv

1. SEPARABILITY | validation and evidence statement

Requirement	Status	Test/result
All ten category contrasts	PASS	40 supported acquisition-history contrasts, all ten pairs in four cohorts
Whole-spectrum difference	PASS	Effect magnitudes and Holm-adjusted permutation evidence reported separately
Wavelength-level difference	PASS	All 360 retained coordinates; signed/absolute effects, support and null uncertainty
Multiple comparison control	PASS	Synchronized max-T over coordinates; Bonferroni over 40 contrasts. Whole-spectrum Holm family separate.
Effects distinct from p-values	PASS	Numerical effect and adjusted evidence shown on separate axes/tables
Site effects	PASS	Every observed site remains visible; weakest effect domain recorded
Year effects	PASS	NIWO 2020/2023 effects retained separately
Atmospheric gaps	PASS	Heatmaps split into disconnected intervals; no bridge lines
Redundancy and regions	PASS	Training correlation and equal-size subset controls; region effects separated
NEON-to-OLI effect retention	PASS	Shared-source effect ratios and uncertainty, not empirical Landsat validation
Unsupported domains marked	PASS	No WREF/YELL 2020, missing atmospheric intervals excluded

WHAT WE KNOW

All ten researcher-category contrasts differ in these acquisition-matched cohorts. Many broad spectral intervals carry adjusted evidence; magnitudes and directions are not equally stable everywhere. Synthetic OLI often retains substantial standardized effect despite reduced spectral resolution.

WHAT WE DO NOT KNOW

Universal differences across ecological regions, physical pure-material distinctions, independently verified annual biological change, and absolute physical reflectance units are not established.

WHAT WOULD FALSIFY THE CURRENT INTERPRETATION

An independently annotated, adequately supported acquisition-matched sample that fails to reproduce the effect, reverses its domain stability, or reveals an unaccounted source/label/scaling artifact would challenge generalization. A documented violation of within-block exchangeability would weaken the present permutation interpretation.

Evidence: tables/design_validation.csv; tables/whole_spectrum_separability.csv; tables/wavelength_effect_stability.csv

2. IDENTIFIABILITY

Scientific question

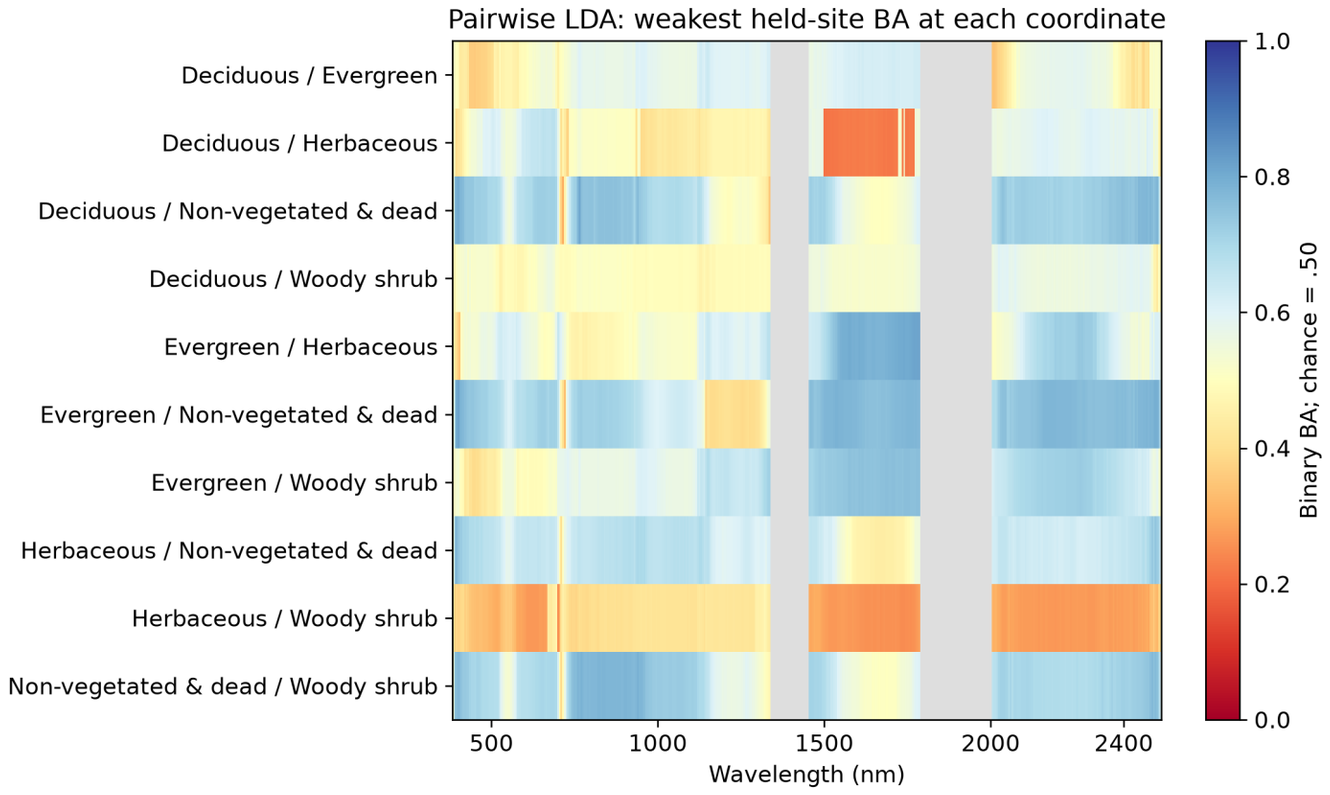
Given an unseen polygon spectrum, can we identify its researcher category? A significant distributional difference does not guarantee a correct label for an individual observation.

Relevant method

Binary comparisons use the prespecified balanced-prior univariate linear discriminant rule: direction and midpoint are learned only from training IDs. Five-class single-coordinate recognition uses nearest training-class mean with equal priors. These rules were already run; they are reused rather than silently replaced with an optimized classifier. Pairwise BA has chance .50, five-class BA .20.

Statistical differences do not guarantee individual recognition





HOW TO READ IT

Left is a five-class problem (chance BA=.20); shaded ranges are observed outer-fold ranges. Right contains ten separate binary problems (chance=.50). No coordinate is chosen using these test scores.

WHAT WE LEARN

Single-coordinate five-class recognition is limited. Herbaceous versus Woody shrub remains especially weak under the fixed pairwise rule despite supported differences.

WHAT REMAINS UNCERTAIN

This tests specified mean-based rules, not all possible nonlinear classifiers. Observed maxima are descriptive and not validated operational bands.

Numerical sources: tables/single_coordinate_five_class.csv; tables/pairwise_single_coordinate_summary.csv

2. IDENTIFIABILITY | complementary information and matched controls

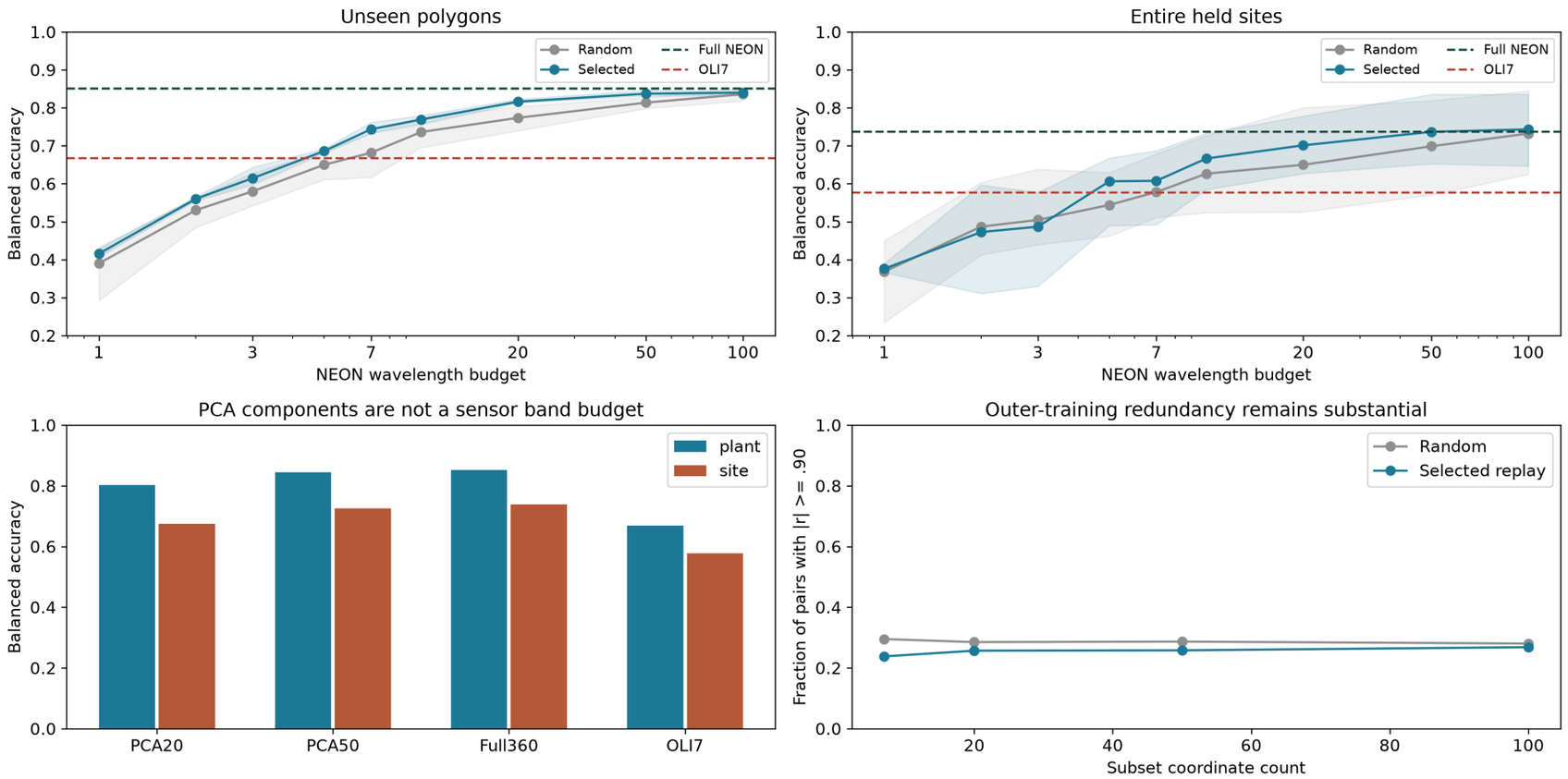
Relevant method

The original greedy orthogonal-residual selector is refitted within each inner group fold; logistic C is chosen from .1/1/10. Five fixed uniform random coordinate permutations now supply nested equal-size controls using the exact same inner folds and C grid. No random seed is chosen by target performance. PCA20/PCA50 and regularized full NEON/OLI provide training-only sensitivities.

Incremental information

At 20 coordinates, new-polygon BA is 81.6% selected versus 77.4% random; held-site BA is 70.1% versus 65.0%. At 100 coordinates the average gap narrows. These curves do not establish an optimal sensor or universal independent information dimension.

Complementary selection adds value, but random bands can catch up



HOW TO READ IT

Selected and random subsets use identical inner folds and tuning grids. Five fixed random paths are averaged; shaded envelopes include seed/fold variation, not confidence intervals. PCA uses every input wavelength.

WHAT WE LEARN

Selected subsets improve modest-budget average recognition. The gap narrows at large budgets; two/three-band held-site controls can favor random subsets. High correlation does not make every additional band useless.

WHAT REMAINS UNCERTAIN

No seed or budget is chosen from outer scores. Five random controls do not establish an optimal selector, universal intrinsic dimension or optimized sensor. PCA fit uses annual rows within training only.

Numerical sources: tables/recognition_control_summary.csv; tables/selected_minus_random.csv; tables/recognition_budget_increments.csv; tables/subset_redundancy.csv; prior:tables/complementary_feature_performance.csv

2. IDENTIFIABILITY | class-specific and domain failures

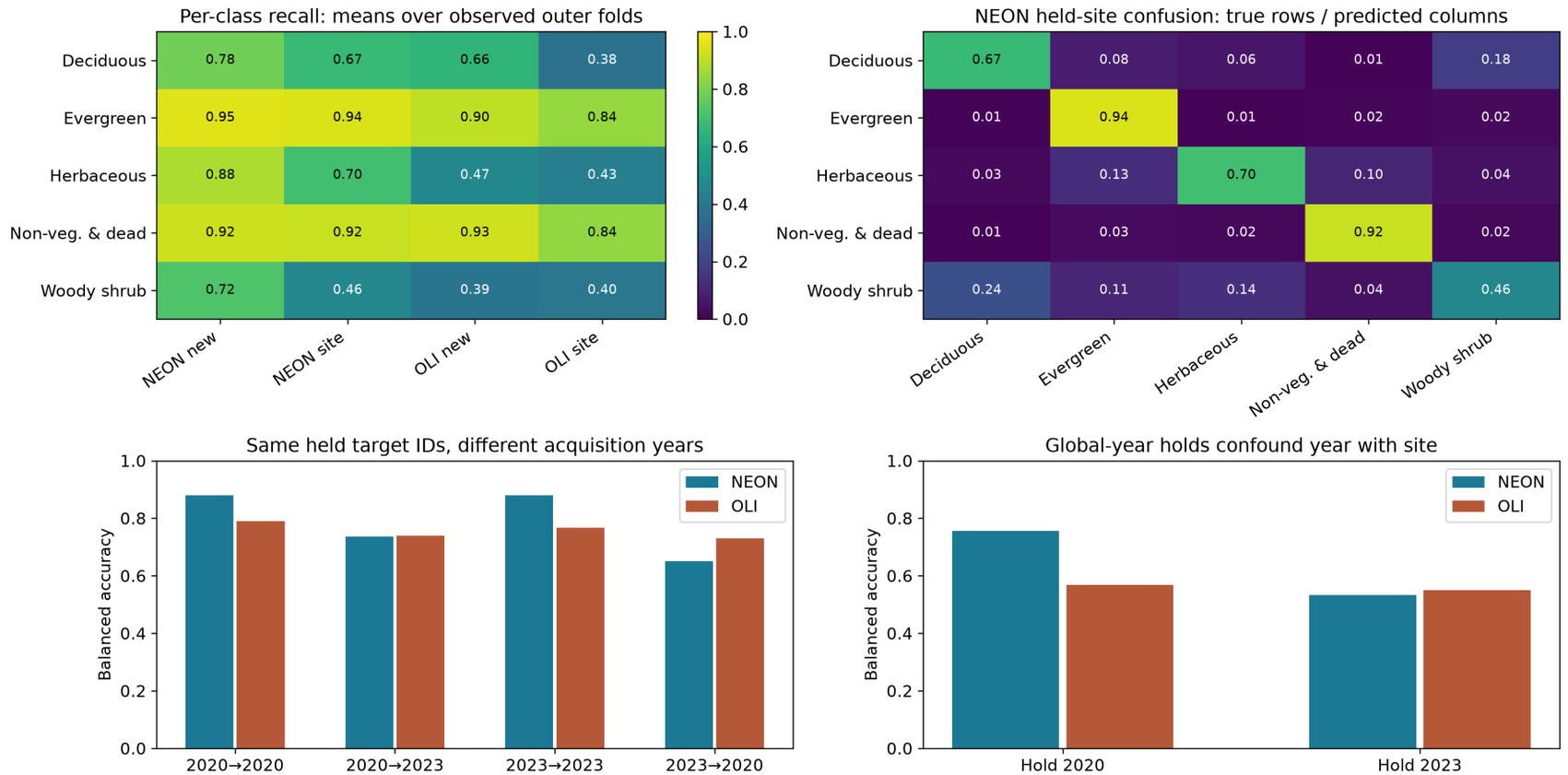
Relevant method

Full-spectrum recognition is identity-balanced regularized logistic with group-aware inner tuning. Every held site is absent from training. The new temporal framework uses the same full/OLI model procedure: global-year holds plus three NIWO held-ID folds in each training-year direction, with the same targets evaluated in both acquisition years.

Transfer penalty

Mean new-polygon to held-site BA falls by 11.4 percentage points for NEON and 9.1 percentage points for OLI. Woody shrub held-site recall is 46.3%. Means alone must not hide this failure or the differing temporal directions.

Recognition failures are class-specific and domain-dependent



HOW TO READ IT

Recall measures recovered true-category observations. Confusion rows sum to one. Paired-year targets keep the same held identities and trained model; global-year targets change site composition.

WHAT WE LEARN

Woody shrub transfers poorly relative to Evergreen/background. Controlled paired NEON recognition drops in both temporal directions and is asymmetric. OLI can be less temporally sensitive despite lower familiar-domain recognition.

WHAT REMAINS UNCERTAIN

Static retrospective annotations test footprint recognition, not annual biological truth. Three sites and one repeatedly observed site cannot isolate ecological, illumination and processing causes.

Numerical sources: prior:tables/class_specific_transfer.csv; prior:tables/confusion_plot_values.csv; tables/temporal_recognition_summary.csv; tables/rates.csv; tables/confusions.csv

2. IDENTIFIABILITY | validation and evidence statement

Requirement	Status	Test/result
Unseen identities	PASS	Outer target GlobalIDs absent from all model training
Nested feature selection	PASS	Prior selector trained fresh within inner folds; target-free random paths
Nested tuning	PASS	C chosen on exact group-aware inner folds
Entire held sites unseen	PASS	Three site holds; poor sites not averaged away in saved scores
All five classes evaluated	PASS	Recall, precision, FPR/FNR and confusion counts per class
Confusion matrices	PASS	Five-class confusion matrices for each full/PCA/random/selected case
Pairwise versus five-class distinguished	PASS	Binary LDA versus balanced-prior five-class nearest-mean rule
Single-coordinate versus full spectrum	PASS	Separate prediction tasks, 50% versus 20% chance BA
Complementary information	PASS	Nested selector feature-budget curve and increments
Equal-size random controls	PASS	Five fixed random paths, same inner folds/C grid; no winning seed selected
Redundancy tested	PASS	Within-subset correlations, random/PCA/full-spectrum sensitivities
Geographic transfer	PASS	All site folds plus class-specific failures
Temporal transfer	PASS	Consistent nested full NEON/OLI; paired identity targets both years
NEON versus OLI predictive loss	PASS	Same new-polygon/site outer framework; temporal losses also retained

WHAT WE CAN IDENTIFY

The five-category label of unseen sampled-population polygons with useful but imperfect accuracy. Evergreen and background transfer more reliably than Woody shrub. Complementary subsets add average moderate-budget value over equal-size random controls.

WHAT WE CANNOT RELIABLY IDENTIFY

Every Woody shrub footprint, every small-subset target, or observations from arbitrary new regions/years. Category recognition is not species identity, material purity or unique mixture composition.

WHERE IDENTIFICATION FAILS

Held sites, asymmetric paired image-year transfer, difficult category overlaps, and insufficient individual-coordinate information. Small selected budgets do not uniformly outperform random subsets.

WHAT ADDITIONAL DATA WOULD MOST IMPROVE IDENTIFICATION

Independently annotated footprints across more sites and acquisition years, deliberate hard-class sampling, spatially separated target plots and contemporaneous repeated biological labels.

Evidence: tables/design_validation.csv; tables/metrics.csv; tables/selected_minus_random.csv; tables/temporal_recognition_summary.csv

3. ABUNDANCE

Scientific question

Given a mixed spectrum, can we recover the fraction belonging to each researcher category? Fraction accuracy is the target. Spectral reconstruction alone is not a biological fraction validation.

Relevant method

Reuse the fixed 170 pair/triple-grid mixtures per five-parent bundle. A training context has 24 proper bundles (120 distinct parents), eight tuning bundles (40 parents) and 12 target bundles (60 parents). GlobalIDs are disjoint across roles; all primary methods receive identical outer targets. Twelve dependent bundles, not 2,040 synthetic rows, define each target experiment.

Models and constraints

No-spectrum proper-training fraction mean, FCLS, EAR/MESMA, ridge, PLS, forest, boosting and two-stage presence/conditional regression are compared. Hyperparameters/scalers/transforms are fitted only in the appropriate training/tuning roles. Supervised fractions use independently tested Euclidean simplex projection; raw error/constraint violations are retained.

Oracle before model ranking

Given the actual generating parents, the ideal linear constrained oracle recovers fractions to near machine precision. That verifies the algebraic solver/mixture control; failure with a candidate library cannot simply be blamed on the solver. The oracle cannot be deployed and does not validate physical mixing or purity.

Uncertainty and domain scope

Paired bootstrap draws resample whole target parent bundles, retaining matching method errors. Report per-target intervals against MESMA and the no-spectrum prior. New-parent, held-site, global-year and complete-pair temporal targets remain distinct; the same identities across paired target years are not independent replicates.

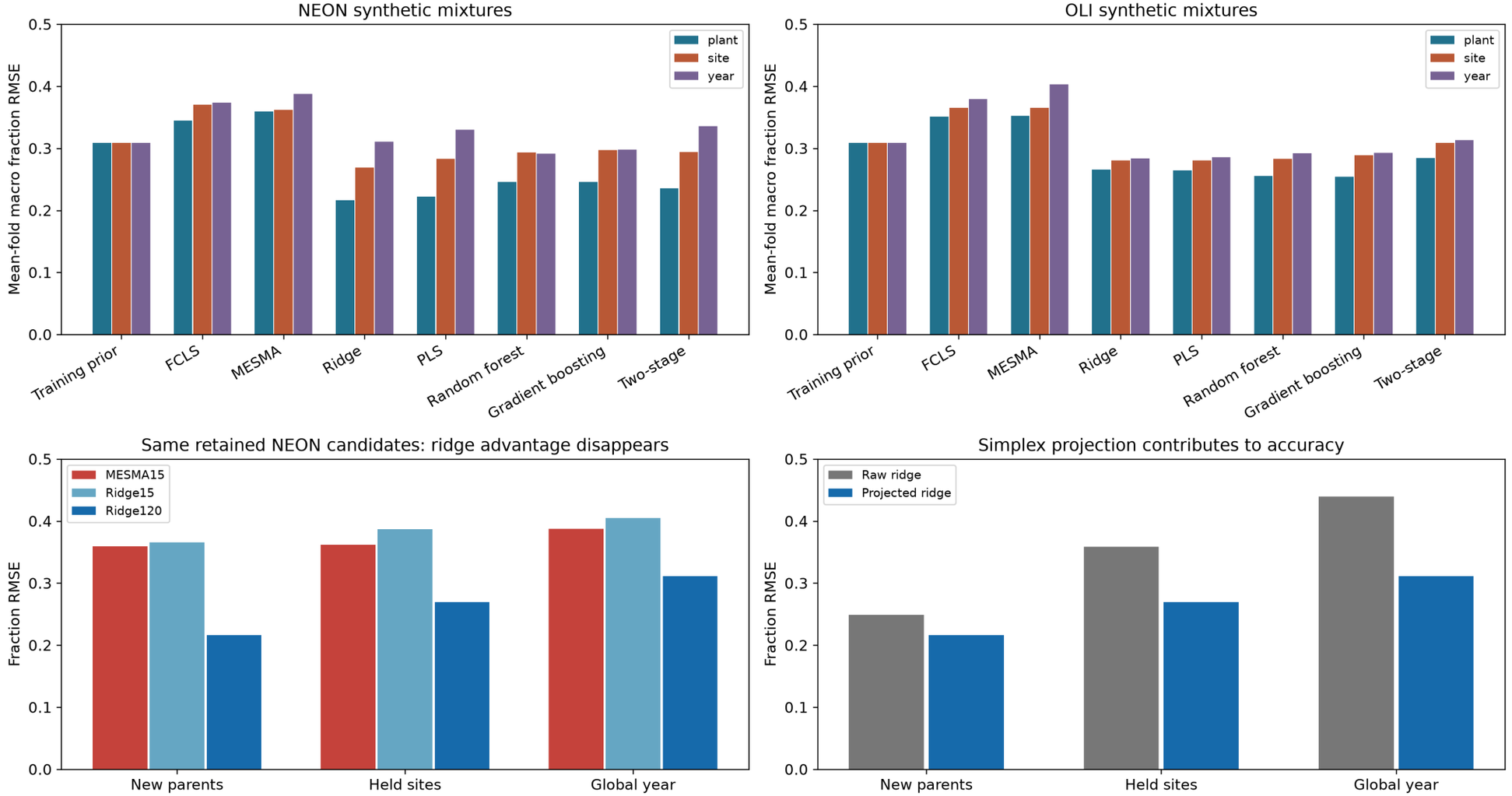
The most important new adverse test

The main ridge uses all 120 proper parents but MESMA retains 15 candidates from that pool. The new fairness control uses exactly those 15 saved MESMA parents to train ridge, with the unchanged tuning/target sets. Its failure materially limits an intrinsic ridge-versus-MESMA interpretation.

Evidence: DESIGN.md; ABUNDANCE_CONTROL_DESIGN.md; prior:METHODS.md; prior:tables/abundance_parent_memberships.csv; prior:tables/oracle_solver_controls.csv; tables/same_candidate_ridge.csv

3. ABUNDANCE | model, prior, projection and capacity controls

Synthetic abundance gains depend on domain, priors and retained library



HOW TO READ IT
All main models see identical outer mixtures. Macro RMSE is sqrt(mean class MSE) within target then averaged across folds. The fairness control reduces ridge to the exact 15 MESMA candidates; the original ridge uses 120 parents.

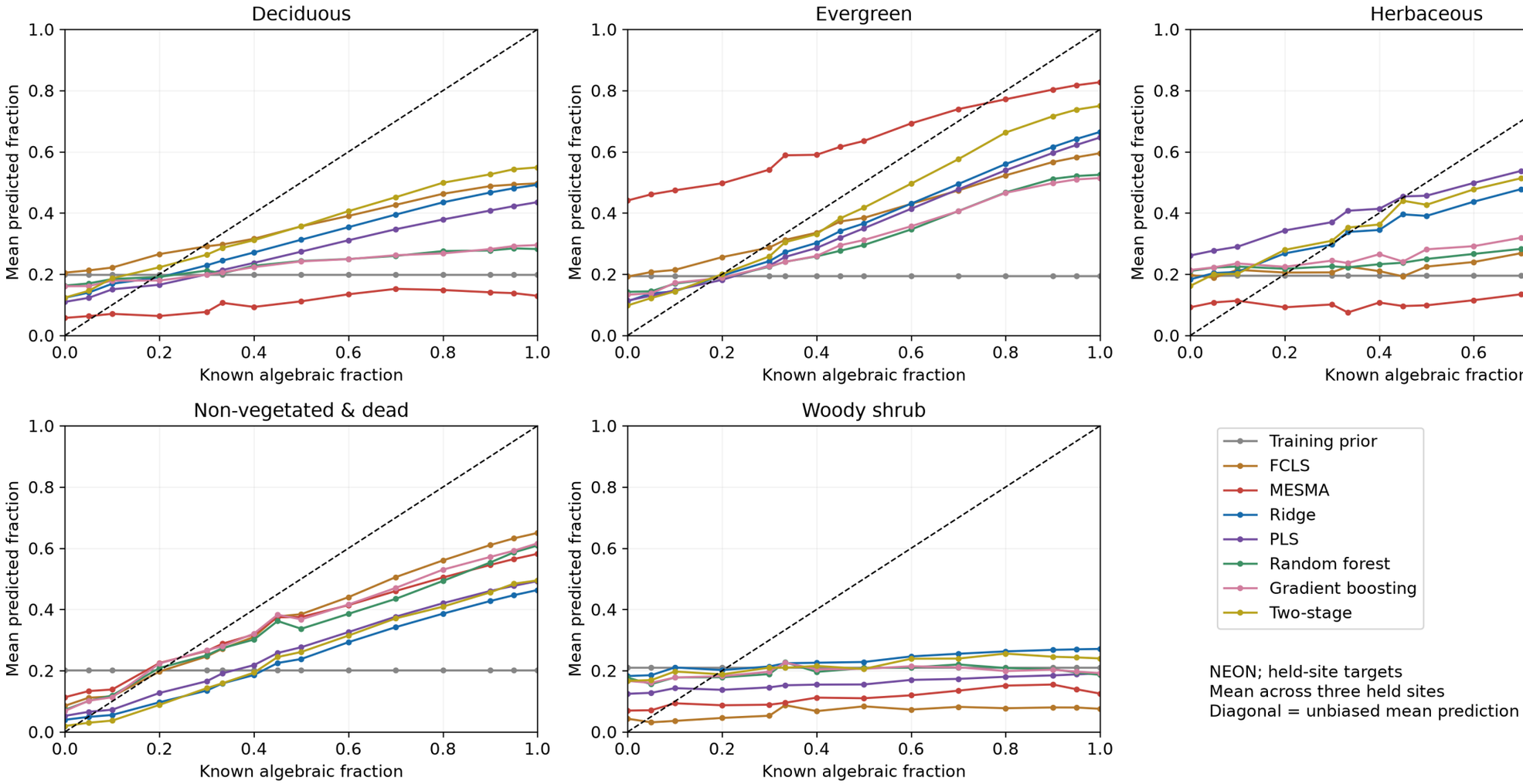
WHAT WE LEARN
Ridge120 improves the primary new-parent/held-site benchmark, but fails the global-year prior on average. Ridge15 loses the NEON advantage. Projection improves error. The known-generating-parent oracle has near-zero fraction error.

WHAT REMAINS UNCERTAIN
The candidate control changes training diversity/bundle count and model prior; it does not prove a single cause. Methods are not intrinsically ranked, and these algebraic fractions are not field cover.

Numerical sources: tables/abundance_domain_summary.csv; tables/same_candidate_ridge.csv; tables/abundance_paired_vs_baselines.csv; prior:tables/raw_prediction_constraint_diagnostics.csv; prior:tables/oracle_solver_controls.csv

3. ABUNDANCE | error across the supported true-fraction grid

Zero, low and pure-end mixtures reveal shrinkage hidden by RMSE



NEON; held-site targets
Mean across three held sites
Diagonal = unbiased mean prediction

HOW TO READ IT
Each point averages predictions at a nominal truth-grid fraction, first count-weighting machine-precision-equivalent bins within fold, then averaging held sites. The diagonal represents unbiased mean recovery. Endpoints are synthetic pure-parent mixtures, not verified pure materials.

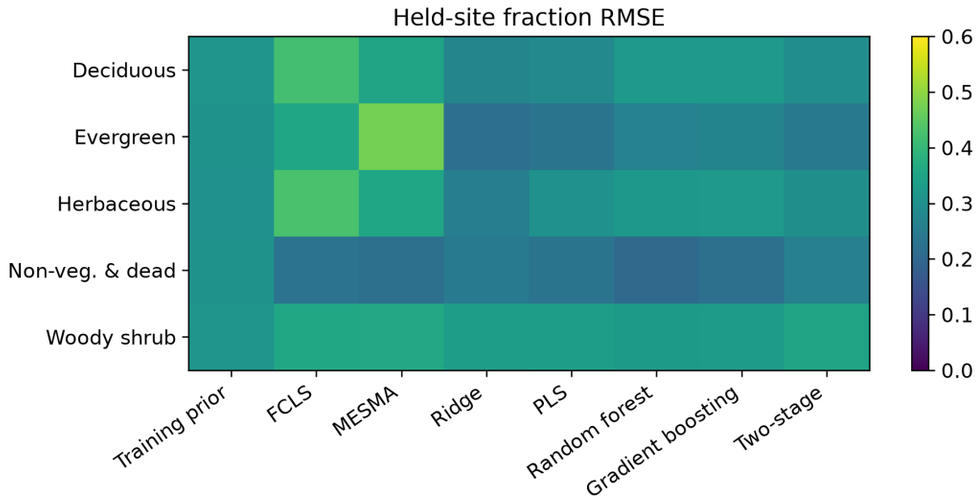
WHAT WE LEARN
Many methods overpredict absent categories and shrink high fractions. Similar overall RMSE can hide different class-specific biases and detection behavior.

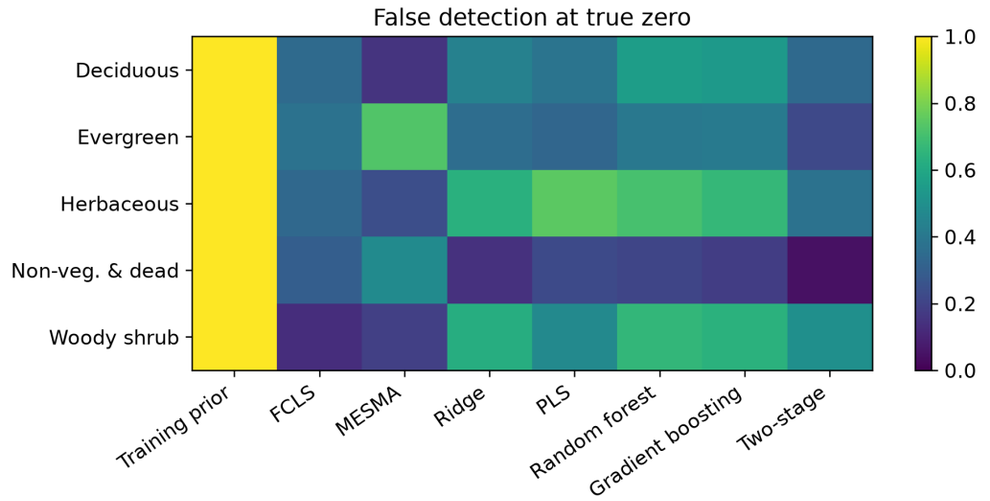
WHAT REMAINS UNCERTAIN
Mean calibration curves hide bundle variation; paired intervals and per-fold/per-class errors remain saved. No universal field detection threshold follows from this grid.

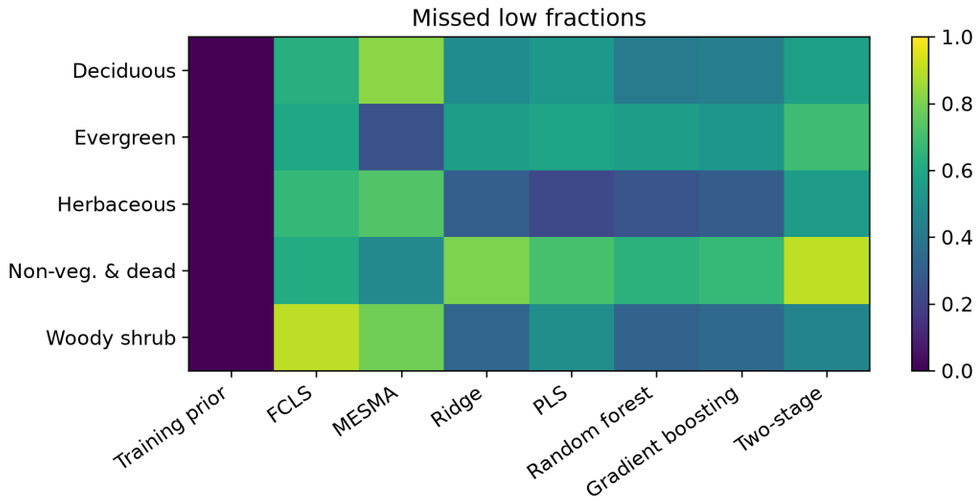
Numerical sources: tables/calibration_plot_values.csv; tables/fraction_grid_calibration.csv

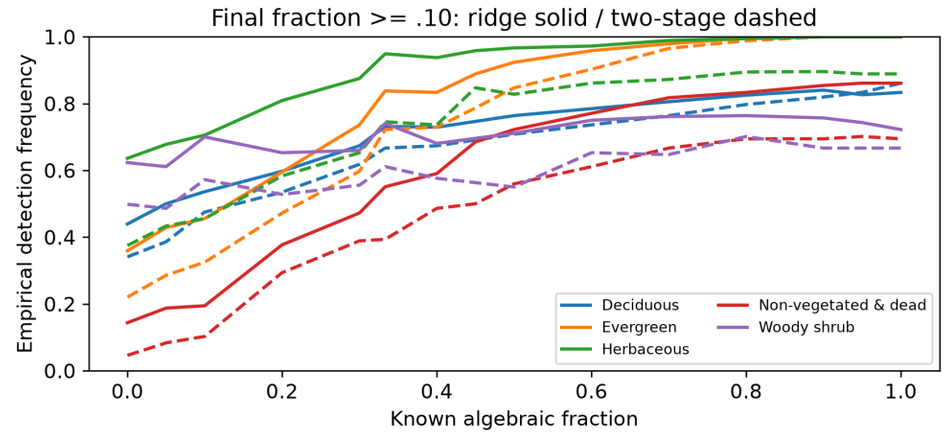
3. ABUNDANCE | presence, amount and category-specific overlap

Recognition, false detection and quantification are different outcomes









HOW TO READ IT

Rows are researcher categories and columns are methods. False detection and low-fraction misses use the declared .10 final-fraction cutoff. The final panel shows empirical detection frequency versus truth, ridge solid and two-stage dashed. Class-level links are saved separately for all five categories.

WHAT WE LEARN

Harder-to-recognize categories also tend to have larger ridge fraction error here. False detection and missed low fractions trade off; a stronger prior can improve one while worsening the other.

WHAT REMAINS UNCERTAIN

Class-level associations are descriptive, not five independent geographic replicates. Stage-A presence metrics are separately saved; .10 is a comparison rule, not an independently validated detection limit.

Numerical sources: prior:tables/abundance_model_comparison.csv; prior:tables/two_stage_presence_metrics.csv; tables/fraction_grid_calibration.csv; tables/class_recognition_abundance_links.csv; tables/class_link_correlations.csv

3. ABUNDANCE | detection and conditional quantification

Two distinct questions

Stage A tests presence using tuning-only thresholds. If every class is rejected, the existing algorithm admits its highest-scoring class; the table reports this admitted gate. Raw versus admitted decisions and detection by fraction are saved separately. Final-fraction detection uses the separate declared .10 comparison cutoff. Positive RMSE below is conditional on true presence, not on the model accepting the category. Missed low fractions and absent-category false detections must be assessed together.

Class-level link

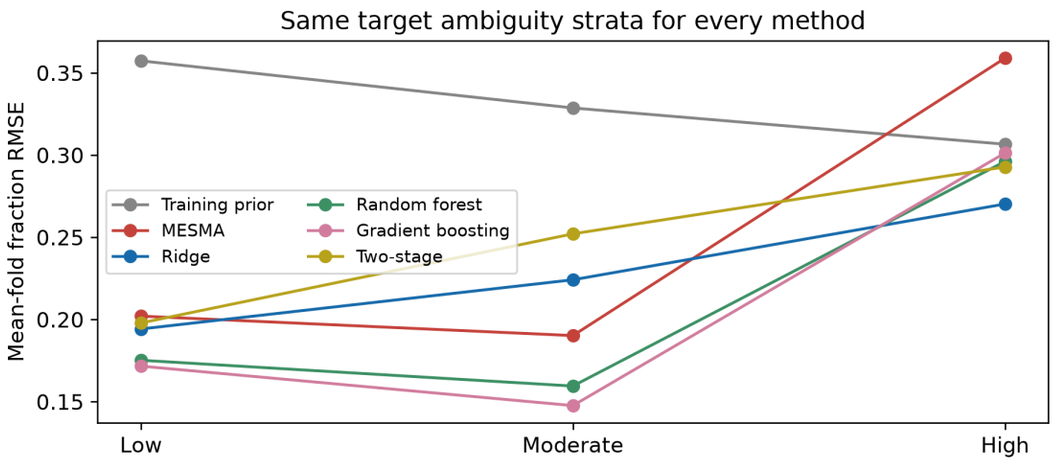
Across these five categories, higher held-site recognition recall is descriptively associated with lower ridge120 fraction RMSE (Spearman rho=-1.0), but low-fraction misses do not follow the same ranking. Five class means do not supply population or causal correlation evidence. Mixture ambiguity burden is the maximum feasible range of any class in mixtures containing the named class, not a class-specific uncertainty interval.

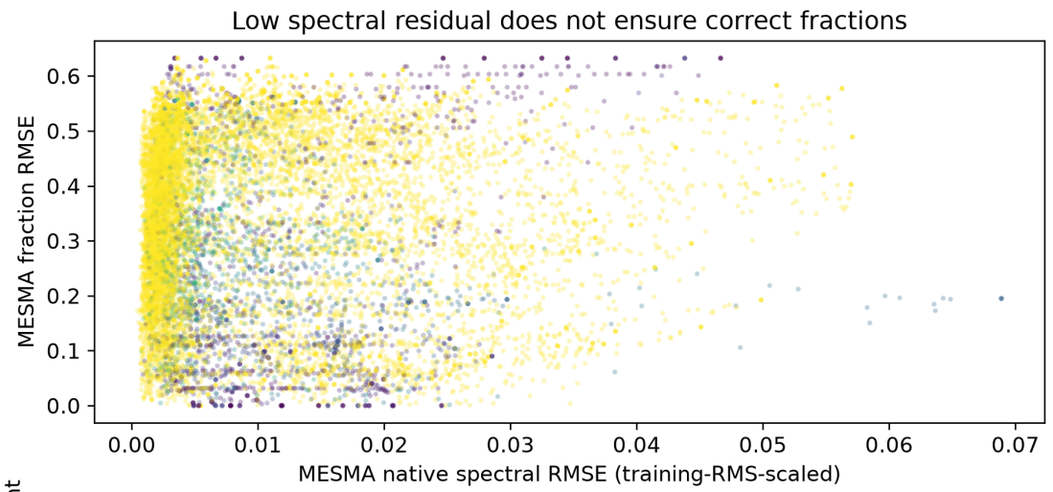
Category	Method	Positive RMSE	Bias	Final FPR	Low FNR	Stage A recall / FPR
Deciduous	Ridge	0.347	+0.000	0.439	0.482	n/a
Deciduous	Two-stage	0.357	+0.018	0.341	0.569	0.691 / 0.386
Evergreen	Ridge	0.246	+0.019	0.359	0.556	n/a
Evergreen	Two-stage	0.269	+0.029	0.220	0.691	0.730 / 0.286
Herbaceous	Ridge	0.268	+0.068	0.636	0.304	n/a
Herbaceous	Two-stage	0.313	+0.064	0.375	0.552	0.737 / 0.395
Non-vegetated & dead	Ridge	0.356	-0.081	0.144	0.808	n/a
Non-vegetated & dead	Two-stage	0.386	-0.089	0.046	0.903	0.581 / 0.091
Woody shrub	Ridge	0.422	-0.006	0.623	0.325	n/a
Woody shrub	Two-stage	0.449	-0.022	0.498	0.452	0.640 / 0.551

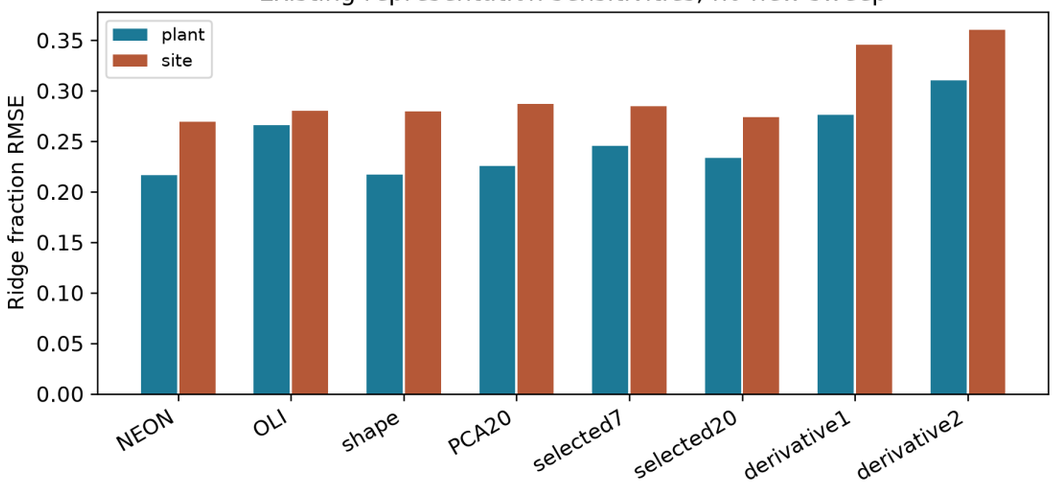
Evidence: prior:tables/abundance_model_comparison.csv; prior:tables/two_stage_presence_metrics.csv; tables/class_recognition_abundance_links.csv; tables/class_link_correlations.csv; tables/stage_A_detection_by_fraction.csv; tables/forced_presence_admission.csv

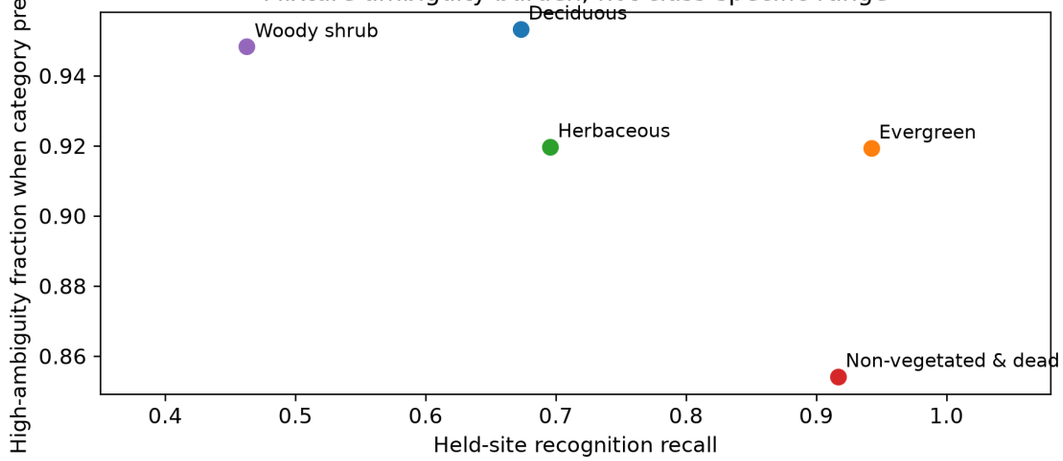
3. ABUNDANCE | ambiguity, reconstruction and spectral representation

Better expected fractions do not establish unique physical inversion









HOW TO READ IT

Ambiguity is maximum candidate-class fraction range among near-fitting solutions at the declared .01 tolerance. High means range >=.30. Every scatter point is one saved synthetic mixture, not an independent plant.

WHAT WE LEARN

Ridge can reduce error in high-ambiguity targets while alternatives remain feasible. The moderate stratum can favor MESMA. Spectral reconstruction is insufficient, and derivatives do not remedy the main limitations.

WHAT REMAINS UNCERTAIN

Candidate/tolerance ambiguity is not exhaustive physical uncertainty. The class burden is mixture-level ambiguity conditional on category presence. Observational data cannot uniquely apportion library variability, overlap, acquisition shift and prior effects.

Numerical sources: prior:tables/abundance_by_ambiguity.csv; tables/reconstruction_fraction_points.parquet; prior:tables/abundance_model_comparison.csv; tables/class_recognition_abundance_links.csv

3. ABUNDANCE | paired uncertainty against both baselines

How to read the table

Each row is a NEON target set. Differences are ridge120 minus the named comparator, so a negative value favors ridge. Brackets are conditional 95% paired whole-bundle bootstrap intervals (999 draws), not joint family-corrected declarations. Target sets from paired years reuse parents; do not pool them as independent biological samples.

Absolute and relative gains

For new-parent targets, mean RMSE decreases 0.143 (39.8%) relative to MESMA; held-site reduction is 0.093 (25.6%). These are gains for the configured unequal-capacity benchmark, not intrinsic method superiority.

Negative controls matter

Global-year mean ridge error 0.311 exceeds the no-spectrum prior 0.309. Same-candidate ridge15 NEON error is 0.365/0.387, so the full-parent gains cannot be generalized to equal retained training diversity.

Training / target	Ridge RMSE	Ridge - MESMA [95% interval]	Ridge - prior [95% interval]
paired_train2020 / target2020	0.219	-0.146 [-0.169, -0.126]	-0.089 [-0.115, -0.066]
paired_train2020 / target2023	0.255	-0.081 [-0.108, -0.052]	-0.053 [-0.080, -0.027]
paired_train2023 / target2020	0.278	-0.099 [-0.128, -0.068]	-0.030 [-0.057, -0.005]
paired_train2023 / target2023	0.199	-0.134 [-0.153, -0.114]	-0.110 [-0.136, -0.086]
plant_0 / target	0.222	-0.145 [-0.172, -0.117]	-0.086 [-0.105, -0.069]
plant_1 / target	0.211	-0.133 [-0.157, -0.109]	-0.097 [-0.121, -0.074]
plant_2 / target	0.215	-0.150 [-0.172, -0.127]	-0.093 [-0.112, -0.075]
site_NIWO / target	0.279	-0.118 [-0.135, -0.098]	-0.030 [-0.046, -0.015]
site_WREF / target	0.272	-0.072 [-0.093, -0.051]	-0.036 [-0.056, -0.016]
site_YELL / target	0.257	-0.089 [-0.120, -0.059]	-0.052 [-0.079, -0.028]
year_2020 / target	0.288	-0.094 [-0.110, -0.079]	-0.020 [-0.036, -0.005]
year_2023 / target	0.334	-0.059 [-0.094, -0.025]	+0.025 [+0.006, +0.043]

Evidence: tables/abundance_paired_vs_baselines.csv; tables/abundance_domain_summary.csv; tables/same_candidate_ridge.csv

3. ABUNDANCE | validation checklist

Requirement	Status	Test/result
Mixture parent separation	PASS	Distinct proper/tuning/target parents; no biological replication from synthetic rows
Grouped dependent mixtures	PASS	Whole-bundle paired uncertainty
Known-parent oracle	PASS	Near machine precision verifies ideal linear solver, not candidate adequacy
Identical targets	PASS	384 canonical prediction cases checked against exact saved truth
Nonnegative sum-one fractions	PASS	Constraints independently verified; raw projection sensitivity retained
Ridge beats MESMA in primary benchmark	PASS	Yes on mean new-parent/held-site NEON under 120-parent ridge versus 15-candidate MESMA
Paired uncertainty	PASS	Conditional bootstrap differences versus MESMA AND training prior
Class-specific errors	PASS	RMSE, MAE, bias, slope, false detection and low-fraction miss per class
True-fraction calibration	PASS	Full supported grid including zero/pure endpoints; shrinkage explicit
False detections	PASS	At true zero, projected fraction >= .10 declared comparison rule
Low-fraction misses	PASS	Conditional detection-versus-fraction and low_FNR
Detection versus quantification	PASS	Stage A presence separate from final constrained fraction errors
Ambiguity calculated	PASS	Every primary target assigned common NEON candidate-library max fraction-range stratum
RMSE versus non-unique fits	PASS	Better expected fraction accuracy coexists with alternative candidate compositions
Held-site abundance	PASS	All three holds, each method against prior
Temporal abundance	PASS	Global-year and complete-pair target comparisons; site and retrospective truth caveats
NEON/OLI/transforms	PASS	Existing full/OLI/shape/PCA/selected/derivative results reused without sweep
Synthetic versus field	PASS	Synthetic algebraic fractions explicitly distinct from field fractional cover
Ridge beats prior across every domain	FAIL	Global-year NEON mean ridge error exceeds no-spectrum prior; not transferable skill in every domain
Ridge superiority after equal retained parent capacity	FAIL	15-parent NEON ridge loses its original advantage over 15-candidate MESMA
Unique physical inversion	NOT TESTABLE	Candidate-library alternatives disprove unique candidate solutions for many targets; exhaustive physical uniqueness unavailable

Evidence: tables/design_validation.csv

3. ABUNDANCE | limiting factors and evidence statement

Potential limit	What the evidence actually supports
Solver failure	Ideal known-parent oracle is near exact. Solver failure is not the explanation for the controlled candidate-library errors.
Spectral resolution	OLI loses recognition and some synthetic fraction information. Shared-source convolution does not test actual satellite scaling or PSF.
Candidate-library variability	Ridge120 gains disappear in the same-candidate NEON ridge15 control. Retained parent diversity/capacity matters, but the control does not isolate one causal mechanism.
Class overlap	Candidate near-solutions and recognition errors persist. Better expected fractions do not make every composition unique.
Geographic shift	Held-site recognition and fraction errors worsen. Three observational sites cannot establish broad geographic universality.
Temporal shift	Paired recognition/abundance differences and global-year failures remain. Static annotations and confounding block ecological causal attribution.
Training priors	The no-spectrum prior defeats ridge on global-year mean. Projection and learned fraction/shape priors contribute to performance.
Independent field truth	Missing. No synthetic result can establish real fractional cover, physical purity or real mixing-linearity validity.

WHAT WE CAN ESTIMATE

Known algebraic researcher-category weights in this finite-parent synthetic benchmark; these are not independently measured field fractions.

HOW ACCURATELY

Full-parent ridge120 mean RMSE is 0.216 for new parents and 0.269 for held sites. Class-specific biases, false detections and low-fraction misses remain substantial.

WHERE IT TRANSFERS

Some sampled held-site and paired acquisition-year targets retain useful conditional skill. This does not establish performance in arbitrary regions or years.

WHERE IT FAILS

Global-year mean fails the no-spectrum prior, same-candidate NEON ridge loses its original advantage, and derivatives worsen the main results. Do not select a winner solely from one favorable mean.

WHETHER THE INVERSION IS UNIQUE

Not established. Alternative candidate compositions can fit nearly equally well even when expected fraction error is reduced. Learned priors can improve prediction without removing non-uniqueness.

WHAT IS STILL REQUIRED FOR FIELD ABUNDANCE

Independent contemporaneous field fractions at aligned spatial support, validated annotation/material purity and reflectance units, real mixing/nonlinearity checks, and broader site/year external targets. More mixtures from the same parents do not fill these gaps.

Evidence: tables/abundance_domain_summary.csv; tables/same_candidate_ridge.csv; prior:tables/abundance_by_ambiguity.csv; prior:tables/abundance_model_comparison.csv

FINAL EVIDENCE LADDER

Level / question	Status	Exact supporting evidence	Scope
1 / SEPARABILITY	ESTABLISHED	All ten researcher-category pairs show blocked whole-spectrum differences in four sampled cohorts; adjusted wavelength evidence retained.	Observed labelled footprints, declared exchangeability/units assumptions; not universal ecological difference.
2 / IDENTIFIABILITY	CONDITIONAL	Nested unseen-ID recognition; weaker held-site and temporal performance; selected/random/PCA controls.	Class-specific errors, sampled sites and retrospective labels; not guaranteed recognition in new regions.
3 / ABUNDANCE	CONDITIONAL	Known algebraic fractions recovered with conditional errors; oracle works. Ridge120 gains fail same-candidate NEON control and global-year prior control.	Finite-parent synthetic mixtures only; learned prior and non-unique solutions remain.
4 / FIELD ABUNDANCE	NOT ESTABLISHED	No independent contemporaneous field fractions or verified pure parent materials.	Requires new biological/physical observations, not more synthetic mixtures.

LIMITATIONS

The dataset was repeatedly explored before these experiments. Nested splitting protects within-experiment tuning, not all design adaptation to known data. Only three sites and one repeatedly sampled site are available; site/year/class/acquisition effects are not fully crossed. Distinct polygons can be spatially correlated. Retrospective annotations do not certify biological persistence or purity.

Do not collapse the ladder

A significant distributional contrast is not reliable individual recognition; recognition is not unique inversion; low synthetic fraction error is not independent field-cover validation. The four levels retain separate statuses and evidence.

Evidence: tables/evidence_ladder.csv; tables/adversarial_validation.csv

WHAT DATA SHOULD BE COLLECTED NEXT

1. Independent field fraction targets

Collect contemporaneous fractional cover for all five categories at explicitly matched footprint/PSF support, including true zeros, small fractions, intermediate and near-pure cases. Hold these plots out from all library selection, model development and calibration. Include repeated human/measurement uncertainty.

2. Verify physical reference meaning

Document reflectance scaling/units, radiometric lineage, spatial support and parent-material/category purity. Synthetic convolution from the same reference cannot establish satellite/drone interchangeability or certify physical endmembers.

3. Separate geography from time

Add balanced category samples across multiple new regions and years, with independently repeated biological annotations and acquisition metadata. Use spatially separated or buffered targets to test residual neighborhood dependence.

4. Target known failures

Oversample independently verified Woody shrub versus Herbaceous/Deciduous overlaps; include varied within-category conditions and mixed backgrounds. Assess absent-category false detections and low-fraction misses independently of the comparison cutoff.

5. Freeze the next validation protocol

Before collecting/evaluating the new targets, fix models, training-library capacity, fraction priors, spatial aggregation, missingness rules, sensor construction, detection/quantification criteria and acceptable uncertainty. Treat current thresholds and data-dependent discoveries as hypothesis-generating.

Evidence: NEXT_DATA_NEEDED.md; tables/adversarial_validation.csv

REPRODUCIBILITY

Canonical inputs and preserved history

The supplied polygon Parquet is read in place. No source spectral dataset was copied. The sealed P129 packet and its previous report remain immutable; its 196 owned files are checked before/after. All new scientific grouping uses cover_category only.

New analysis protocol

DESIGN.md fixes random/PCA/temporal controls; ABUNDANCE_CONTROL_DESIGN.md records the later hostile-review fairness addition before its evaluation. Fixed seeds, split memberships, feature paths, inner scores, model parameters, target contributions and per-case hashes are saved. This is a documented exploratory reanalysis, not retrospective preregistration.

Numerical reproduction

run_controls.py and run_candidate_control.py are restart-safe. validate_controls.py independently reconstructs all saved classification predictions and inner groups; validate_candidates.py uses independent ridge normal equations. synthesize.py checks every canonical abundance evaluation against saved truth; no model is accepted solely from a summary CSV.

Figures and report

Ten figure families have PNG/PDF/SVG exports. Every caption states how to read it, what is learned and what remains uncertain. figure_manifest.json lists exact numerical sources/hashes. report_source.json and report_source.md are editable; build_report.py renders from saved sources without refitting. Final geometry, rendered-page review, source hashes and deterministic rebuilding are recorded.

Interpretive traceability

tables/design_validation.csv contains every requirement and its PASS/FAIL/NOT TESTABLE result. tables/adversarial_validation.csv records the hostile review. tables/claim_registry.csv maps key quantitative claims to their exact source filters. tables/evidence_ladder.csv keeps four scientific evidence levels separate.

Evidence: README.md; VALIDATION.md; control_validation.json; candidate_validation.json; synthesis_validation.json; figure_manifest.json; tables/claim_registry.csv

Adversarial validation appendix | 1-7

Hostile-review rule

Each major claim must survive its plausible alternative explanations. FAIL explicitly preserves a failed interpretation; it does not disappear because other tests passed. NOT TESTABLE identifies missing evidence rather than an invented substitute.

Claim / failure mode	Test performed	Result / status	Remaining limitation
Separability / Pixel/wavelength pseudoreplication	Identity summaries, acquisition-history blocks and whole-curve shuffles	3,639 IDs; 360 coordinates are not replicates PASS	Labels exchangeable within declared blocks is an assumption
Separability / Multiple testing and correlated wavelengths	Synchronized max-T then 40-contrast Bonferroni; whole-spectrum Holm	All ten pairs supported in four observed cohorts PASS	Not an effect-size confidence band or independent replication
Separability / One site/year drives the effect	Retain all cohort effects/signs/ranks/weakest domain	Magnitude and direction heterogeneity retained PASS	Only NIWO repeats in time; not broad regional sampling
Separability / Physical scaling or annotation errors	Inspect source metadata/validity and annotation dates; reuse shape sensitivity	Relative numerical evidence available; absolute units/annual truth unresolved NOT TESTABLE	No independent unit/purity certification
Identifiability / Repeated-identity or exact-copy leakage	Replay memberships and exact duplicate annual vectors	No shared target IDs; no duplicate annual vectors PASS	Spatial autocorrelation can remain between distinct polygons
Identifiability / Feature/tuning leakage	Recreate original inner group folds, random paths and chosen C; replay all predictions	352 models / 364 target evaluations reconstructed PASS	Dataset explored before this study
Identifiability / Class imbalance/easy background hides poor classes	Balanced training weights and per-class confusion/recall	Woody shrub transfer remains substantially weaker PASS	Broad heterogeneous category definitions remain

Evidence: tables/adversarial_validation.csv

Adversarial validation appendix | 8-14

Hostile-review rule

Each major claim must survive its plausible alternative explanations. FAIL explicitly preserves a failed interpretation; it does not disappear because other tests passed. NOT TESTABLE identifies missing evidence rather than an invented substitute.

Claim / failure mode	Test performed	Result / status	Remaining limitation
Identifiability / Selector adds no value over chance subsets	Five random paths, same budgets/folds/C grid	Moderate-budget benefit; very small subsets not uniformly better; large-budget gap small PASS	Five Monte Carlo controls; no globally optimal bands
Identifiability / Neighbor redundancy mistaken for independent information	Within-subset |r|, PCA20/50, random and full regularized models	Many redundant coordinates; useful complementary combinations still improve accuracy PASS	No universal intrinsic dimensionality identified
Identifiability / Site or acquisition memorization	Hold whole sites; paired same-target-ID years	Geographic and asymmetric temporal penalties remain PASS	Retrospective footprint recognition, not annual ecological change
Abundance / Parent leakage/dependent synthetic rows	Verify proper/tuning/target IDs; common truth and bundle bootstrap	All canonical cases pass; bundles are uncertainty unit PASS	Finite parent set and designed mixtures
Abundance / Constrained solver broken	Known generating-parent oracle	Fraction RMSE near machine precision PASS	Ideal linear control unavailable operationally
Abundance / Ridge simply predicts class prior	Compare every primary model/domain with proper-training fraction mean	Ridge fails global-year mean no-spectrum control FAIL	No transferable spectral skill in every observed domain
Abundance / Unequal retained library makes ridge comparison intrinsic	Refit ridge from exact 15 MESMA candidate parents	NEON advantage disappears in reduced-parent control FAIL	Different priors/hypothesis classes and training diversity prevent causal apportionment

Evidence: tables/adversarial_validation.csv

Adversarial validation appendix | 15-19

Hostile-review rule

Each major claim must survive its plausible alternative explanations. FAIL explicitly preserves a failed interpretation; it does not disappear because other tests passed. NOT TESTABLE identifies missing evidence rather than an invented substitute.

Claim / failure mode	Test performed	Result / status	Remaining limitation
Abundance / Simplex projection creates all gains	Raw versus projected ridge fraction error and independent projection tests	Projection helps, especially shifted domains; raw new-parent ridge still improves on MESMA PASS	Raw fractions are not physically constrained compositions
Abundance / Good reconstruction implies correct fractions	Plot native residual against fraction error, retain ambiguity strata	Low residual can coexist with wrong class fractions FAIL	Reconstruction alone cannot validate biological fractions
Abundance / Lower expected error removes spectral ambiguity	Common candidate-library alternative-composition diagnostic	Improved expected error coexists with high ambiguity FAIL	Not an exhaustive physical uncertainty set
Abundance / Recognition success guarantees fraction recovery	Class-level recall/error/detection/ambiguity links	Descriptive associations only five categories; miss/false-positive tradeoffs differ PASS	No causal or population correlation inference
Field abundance / Synthetic fractions equal real cover	Seek independent field fraction and purity truth	Required observations absent NOT TESTABLE	Acquisition, spatial support, nonlinearity and label purity unvalidated

Evidence: tables/adversarial_validation.csv